

BLINDER TUNNEL

MSBuild launches a masqueraded RuntimeBroker from a user-writable path

HUNTING / MEDIUM CONFIDENCE



MICROSOFT DEFENDER XDR



T1127 · T1036



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EXECUTIVE ASSESSMENT

Decision

- Unit 42 documented a weaponized Visual Studio project that overrides GetFrameworkPaths during design-time evaluation.

- msbuild.exe copies and launches a renamed Microsoft vshost binary as RuntimeBroker.exe under %LOCALAPPDATA%\Microsoft\RuntimeBrokers.

- The execution shape is observable and specific, but the directory and filename are mutable campaign artifacts. Classification remains Hunting.

- Operational value: 8/10. No alerting until local developer workflows are baselined.

THREAT MECHANICS

Observed chain

1 A tailored .csproj opens in Visual Studio and triggers a design-time build before compilation.



2 The malicious target creates the RuntimeBrokers directory and copies staged binaries.



3 msbuild.exe launches RuntimeBroker.exe, a renamed signed vshost.exe.



4 RuntimeBroker.exe.config performs AppDomainManager hijacking and disables ETW.



5 RuntimeBroker.dll is sideloaded; ShelbyLoader V2 establishes persistence and GitHub-based C2.



TELEMETRY CONTRACT

Required evidence

Required: Microsoft Defender for Endpoint process creation telemetry in DeviceProcessEvents.



Documented fields: Timestamp, DeviceName, FileName, FolderPath, ProcessCommandLine, SHA1, InitiatingProcessFileName, initiating paths and stable process identifiers.



Optional enrichment: DeviceFileCertificateInfo, file creation events, registry persistence and network telemetry.



Unavailable: .csproj contents, DLL-load proof and ETW configuration content are not inferred by this query.



DETECTION LOGIC

Primary hunting query

DeviceProcessEvents

| where FileName =~ "RuntimeBroker.exe"

| where FolderPath contains @"\AppData\Local\Microsoft\RuntimeBrokers\"

| where InitiatingProcessFileName =~ "msbuild.exe"

| project Timestamp, DeviceName, AccountUpn, FolderPath,

ProcessCommandLine, SHA1, ProcessVersionInfoOriginalFileName,

InitiatingProcessFileName, InitiatingProcessFolderPath

| order by Timestamp desc

TRIAGE

Investigation path

- Confirm the full path and whether the parent is genuine msbuild.exe from an approved Visual Studio installation.
- Retrieve SHA1 and OriginalFileName; determine whether RuntimeBroker.exe is a renamed vshost binary.
- Inspect adjacent RuntimeBroker.exe.config and RuntimeBroker.dll; look for appDomainManager and etwEnable directives.
- Review Run-key persistence value MicrosoftRuntime and GitHub API connections from the same process lineage.
- Escalate only when provenance, path and adjacent artifacts cannot be explained by approved development tooling.

FALSE POSITIVES & VALIDATION

Promotion gates

- Possible overlap: internal build tooling or developer tests that deliberately copy binaries to LocalAppData.

- Tune by approved repository, signer/hash, developer device group and known automation owner - never by filename alone.

- Success: a controlled reproduction produces the expected parent-child event with populated path and stable IDs.

- Reject: normal enterprise workflows use the same path and parent at material volume without stronger enrichment.

- Failure: process telemetry omits the initiating process or normalizes away the complete path.

CONCLUSION & REFERENCES

Analytical conclusion

- This hunt preserves the strongest directly observed endpoint boundary: trusted developer utility execution of a masqueraded binary from a user-writable directory.

- It does not claim exploit success, DLL sideloading, C2, impact or Iranian attribution.

- Primary: Unit 42, Blinder Tunnel Campaign Targets Iraqi Infrastructure, 2026-10-06.

- Schema: Microsoft Learn, DeviceProcessEvents, accessed 2026-10-07.

- ATT&CK: T1127 Trusted Developer Utilities Proxy Execution; T1036 Masquerading.

- Status: Hunting query - Untested implementation sketch.